PHIVOLCS Special Report – Ms6.5 15 km Northeast of Ormoc City (July 6, 2017)

Epicenter: Around 11.145 °N, 124.724°E (coordinates of Ormoc City) – exact values were not reported

Reported Earthquake Intensity (PEIS)	Province	City/Municipality
VIII (very destructive)	Leyte	Along the surface rupture area in Brgys. Mahawan, Rizal, Hiloctogon, Lim-ao, Tongonan, Kananga and Brgys. Tongonan, Milagro, Lake Danao and Gaas, Ormoc City

Intensity VIII

Places	**Coordinates**
Mahawan, Leyte	11.2183° N, 124.6137° E
Rizal, Leyte	11.1824° N, 124.5916° E
Hiloctogon, Leyte	11.1924° N, 124.661 ° E
Lim-ao, Leyte	11.1601° N, 124.6274 ° E
Tongonan, Leyte	11.1339° N, 124.6110° E
Kananga, Leyte	11.1604° N, 124.5750° E
Milagro, Leyte	11.1045° N, 124.6691° E
Lake Danao, Leyte	11.0726° N, 124.7014° E
Gaas, Leyte	11.0323° N, 124.7409° E
Ormoc City Leyte	11.0384° N, 124.6193° E

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): (8.040370°, 121.554818°)

Distance to PHIVOLCS Measured Epicenter: 491.020 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	4.410895	487.126
P2	4.369714	482.577
P3	4.425343	488.649
P4	4.378753	483.518
P5	4.348583	480.186
P6	4.342360	479.558
P7	4.368941	482.347
P8	4.369828	482.387
P9	4.370671	482.408
P10	4.287101	473.317

Minimum R: 4.287101° (~473.317 km)

Maximum R: 4.425343° (~488.649 km)

Mean R_prox: 4.367219°

Radius std: 0.035821°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): (11.093065°, 124.623635°)

Distance to PHIVOLCS Measured Epicenter: 13.948 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.125629	13.968
P3	0.106130	11.774
P4	0.067141	7.465
P5	0.042745	4.745
P6	0.083062	9.177
P8	0.080413	8.786
P10	0.054837	6.097

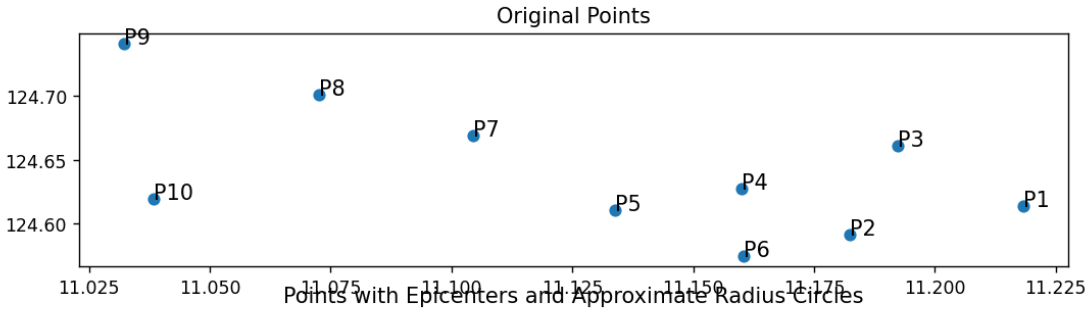
Minimum R: 0.042745° (~4.745 km)

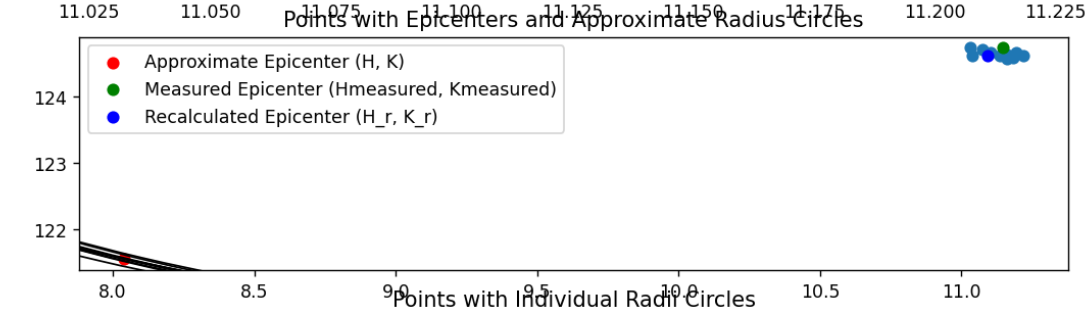
Maximum R: 0.125629° (~13.968 km)

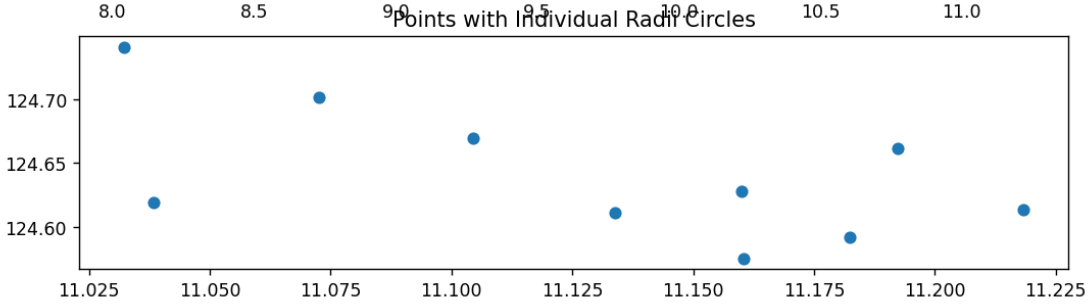
Mean R_prox: 0.079994°

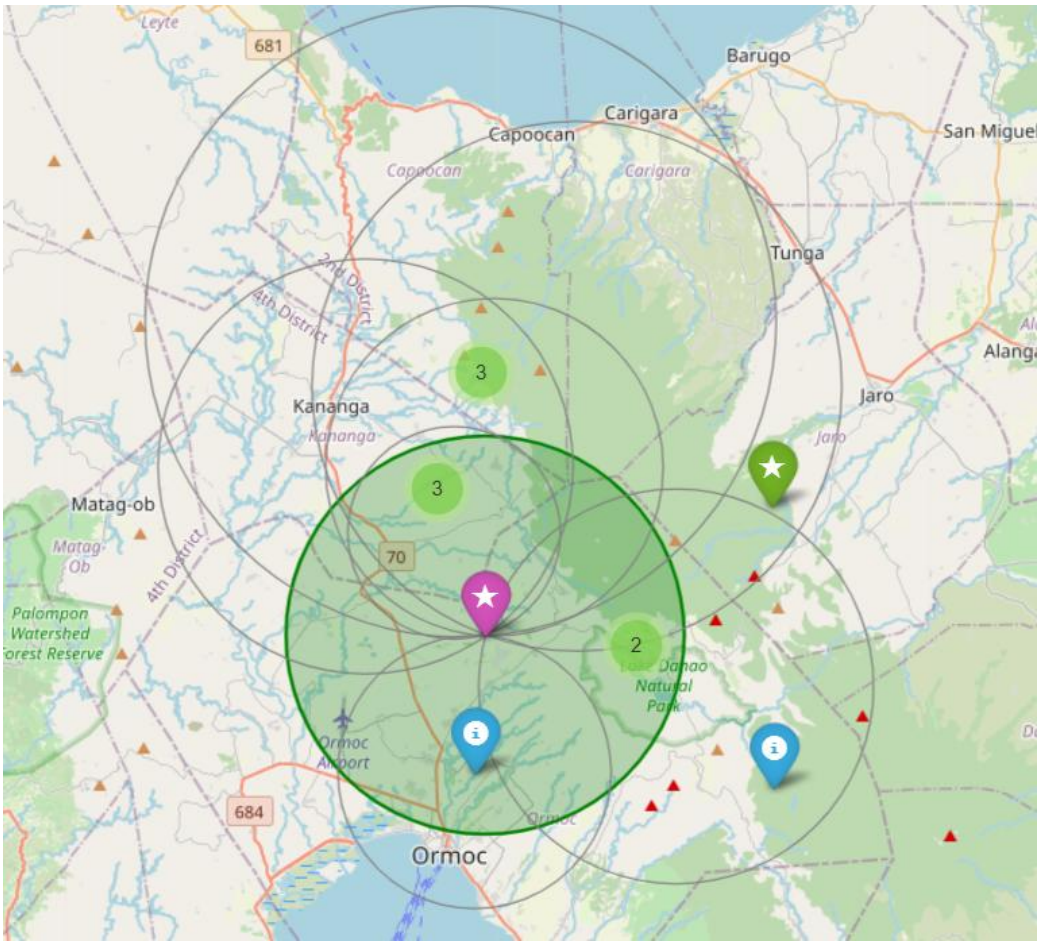
Radius std: 0.026621°

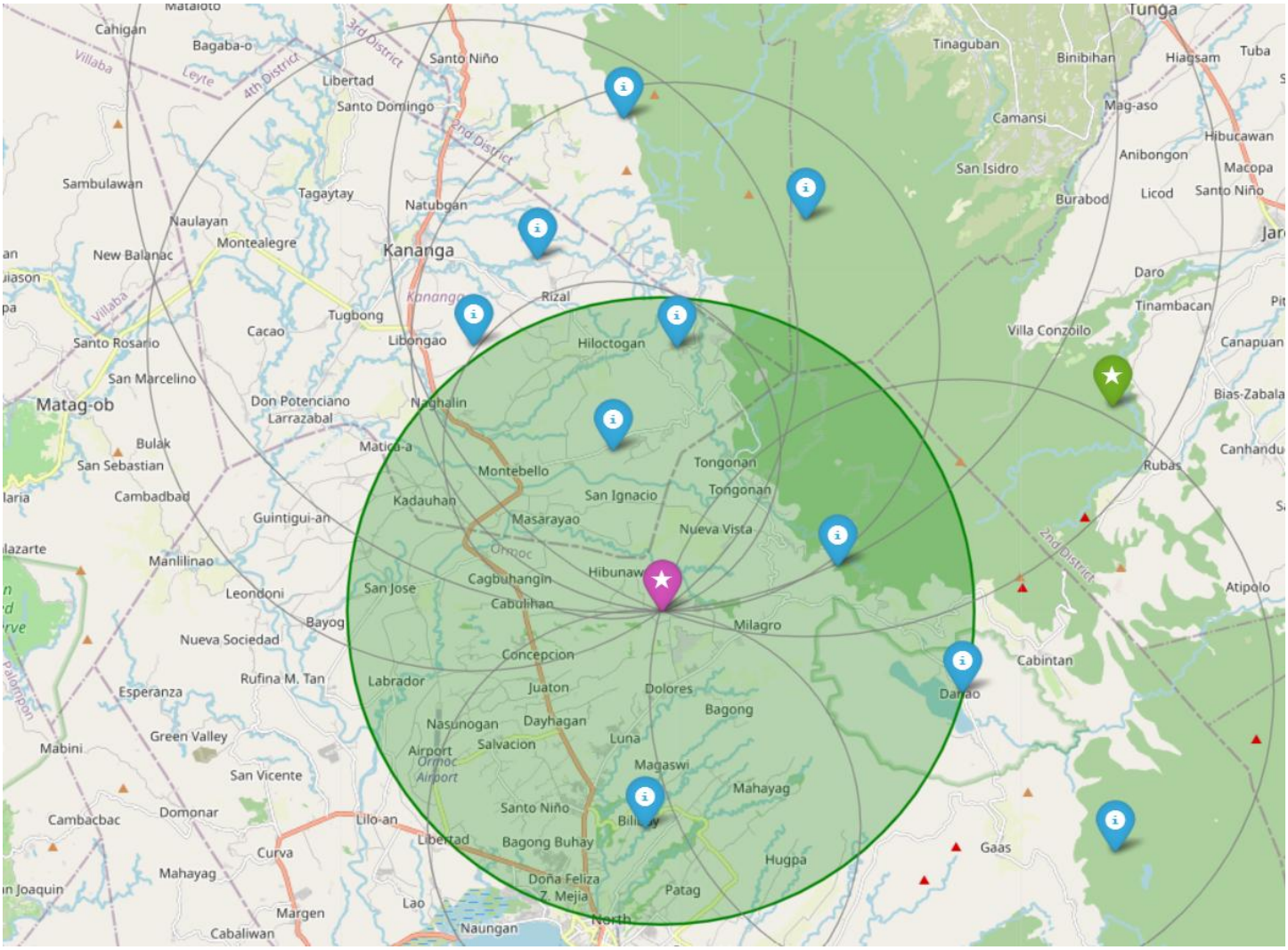
Outliers: One calculation of these points (Rizal Leyte, Milagro Leyte, and Gaas Leyte) produced an outlier.


Original Points
P9
P8
P7
P3
P10
P4
P5
P1
P2
P6
124.70
124.65
124.60
11.025
11.050
11.075
11.100
11.125
11.150
11.175
11.200
11.225


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
124
123
122
8.0
8.5
9.0
9.5
10.0
10.5
11.0


Points with Individual Radii Circles
124.70
124.65
124.60
11.025
11.050
11.075
11.100
11.125
11.150
11.175
11.200
11.225


Leyte
681
Barugo
Carigara
Capoocan
San Miguel
Tunga
Alanga
Jaro
Kananga
3
3
70
2
Matag-ob
Palompon
Watershed
Forest Reserve
Danao
Natural
Park
Ormoc
Airport
684
Ormoc



Blue: Affected intensity points, **Pink:** Recalculated Epicenter, **Green:** Instrumentally reported Epicenter

Intensity VII

VII (destructive)	Leyte	Capoocan and Rest of Ormoc City and Kananga

Places	Coordinates	Findings (from the Method)
Capoocan, Leyte	11.2709° N, 124.6193° E	
Ormoc City	11.0384° N, 124.6193° E	
Kananga	11.1604° N, 124.5750° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): (11.154650°, 124.749306°)

Distance to PHIVOLCS Measured Epicenter: 2.962 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.174401	19.188
P2	0.174401	19.192
P3	0.174401	19.026

Minimum R: 0.174401° (~19.192 km)

Maximum R: 0.174401° (~19.188 km)

Mean R_prox: 0.174401°

Radius std: 0.000000°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): (11.154650°, 124.749306°)

Distance to PHIVOLCS Measured Epicenter: 2.962 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.174401	19.188
P2	0.174401	19.192
P3	0.174401	19.026

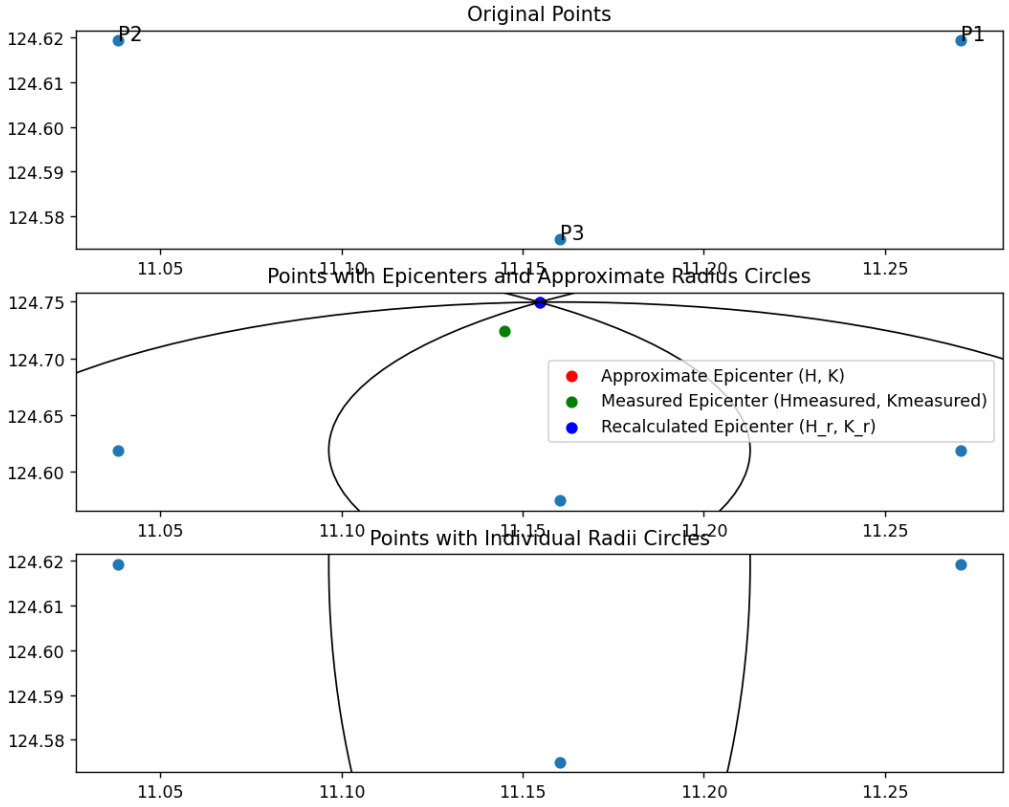
Minimum R: 0.174401° (~19.192 km)

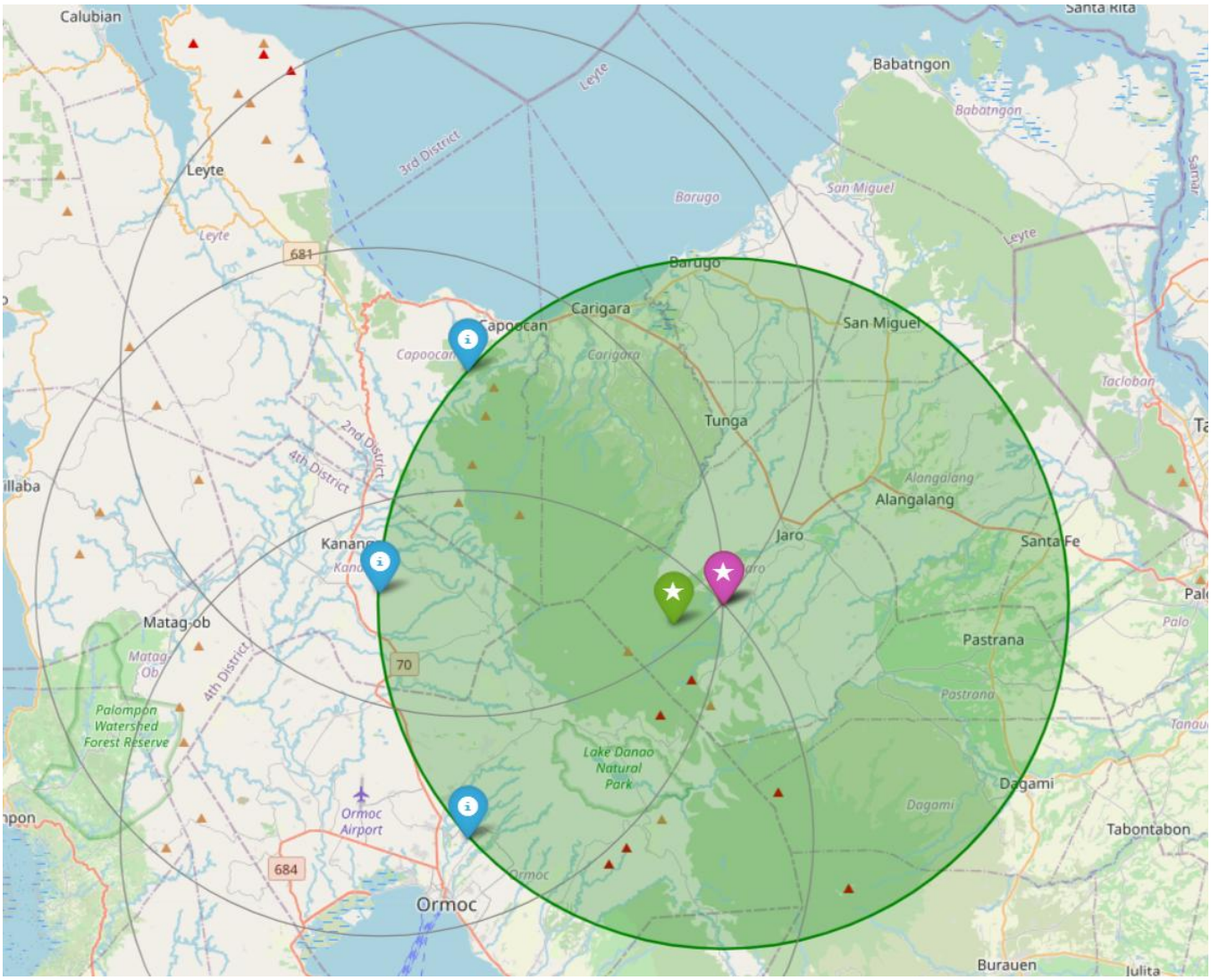
Maximum R: 0.174401° (~19.188 km)

Mean R_prox: 0.174401°

Radius std: 0.000000°

Outliers: None detected





Blue: Affected intensity points, **Pink:** Recalculated Epicenter, **Green:** Instrumentally reported Epicenter

Intensity VI

VI (very strong)	Leyte	Calubian, San Isidro, Tabango, Leyte, Villaba, Matag-ob, Palonpon, Isabel, Merida, Carigara, Barugo, San Miguel, Alangalang, Pastrana, Jaro, Dagami, Burauen, Julita, Albuera

Places	Coordinates	Findings (from the Method)
Calubian, Leyte	11.4670° N, 124.3865° E	
San Isidro, Leyte	11.4310° N, 124.3643° E	
Tabango, Leyte	11.3390° N, 124.3976° E	
Villaba, Leyte	11.1777° N, 124.4420° E	
Matag-ob, Leyte	11.1443° N, 124.4753° E	
Palonpon, Leyte	11.0337° N, 124.4309° E	
Isabel, Leyte	10.9341° N, 124.4531° E	
Merida, Leyte	10.9643° N, 124.5196° E	
Carigara, Leyte	11.2591° N, 124.7078° E	
Barugo, Leyte	11.3002° N, 124.7630° E	
San Miguel, Leyte	11.3078° N, 124.8514° E	
Alangalang, Leyte	11.2356° N, 124.8845° E	
Pastrana, Leyte	11.1197° N, 124.8845° E	
Jaro, Leyte	11.1758° N, 124.7520° E	
Dagami, Leyte	11.0483° N, 124.8403° E	
Burauen, Leyte	10.9711° N, 124.8403° E	
Julita, Leyte	10.9842° N, 124.9617° E	
Albuera, Leyte	10.9000° N, 124.7188° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): (10.015787°, 123.423468°)

Distance to PHIVOLCS Measured Epicenter: 189.666 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	1.741680	192.633
P2	1.699410	187.959
P3	1.643115	181.599
P4	1.545137	170.543
P5	1.542690	170.189
P6	1.432155	157.927
P7	1.379653	151.991
P8	1.449546	159.646
P9	1.787550	197.009
P10	1.855819	204.508
P11	1.925692	212.100
P12	1.903302	209.504
P13	1.831185	201.425
P14	1.763697	194.222
P15	1.753139	192.800
P16	1.708811	187.831
P17	1.817686	199.698
P18	1.568349	172.420

Minimum R: 1.379653° (~151.991 km)

Maximum R: 1.925692° (~212.100 km)

Mean R_prox: 1.686034°

Radius std: 0.161046°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): (11.220905°, 124.632344°)

Distance to PHIVOLCS Measured Epicenter: 13.084 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.347854	38.304
P2	0.340570	37.415
P3	0.262776	28.770
P6	0.275001	30.272
P7	0.338209	37.412
P8	0.280281	31.072
P9	0.084572	9.261
P10	0.152835	16.756
P11	0.235661	25.769
P12	0.252584	27.550
P13	0.271707	29.720
P14	0.127875	13.982
P15	0.270255	29.717
P16	0.325035	35.867
P17	0.405591	44.545

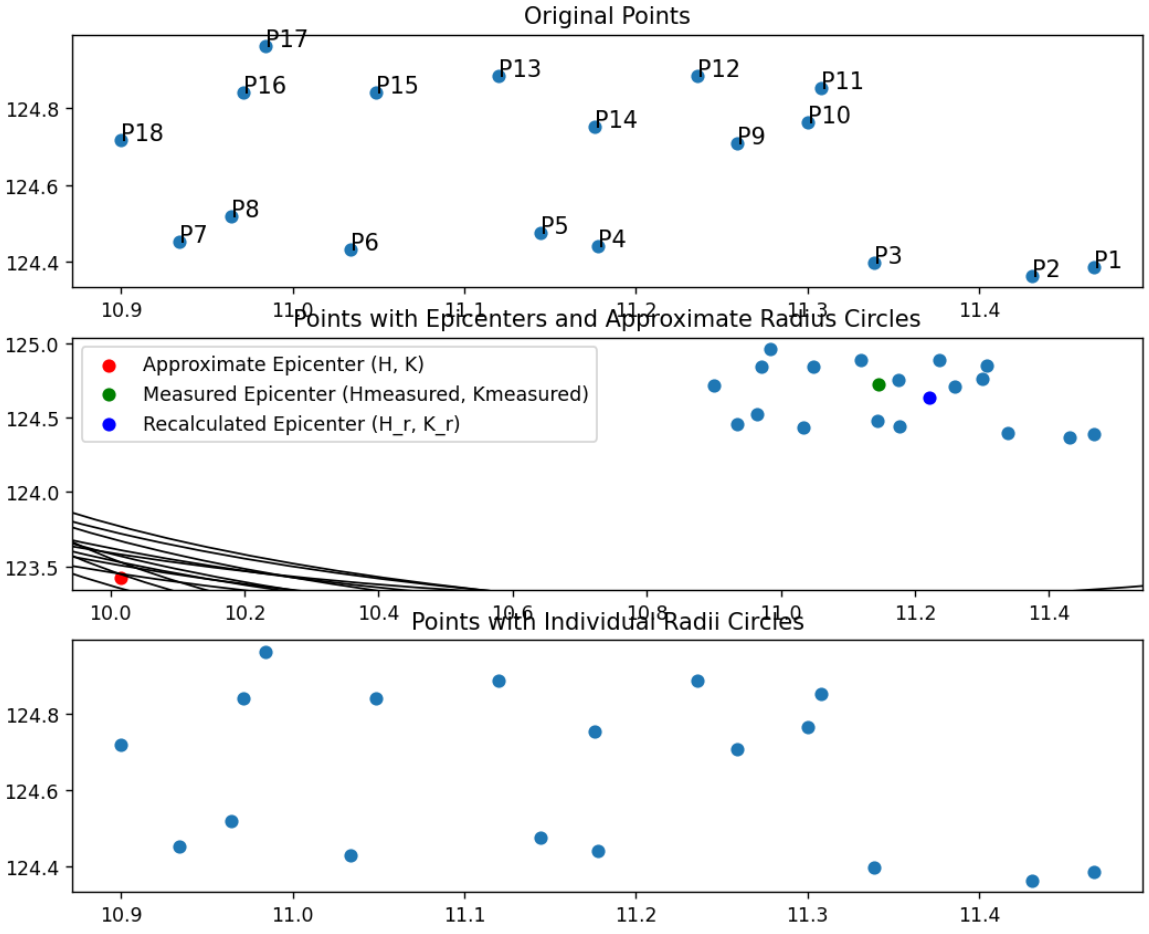
Minimum R: 0.084572° (~9.261 km)

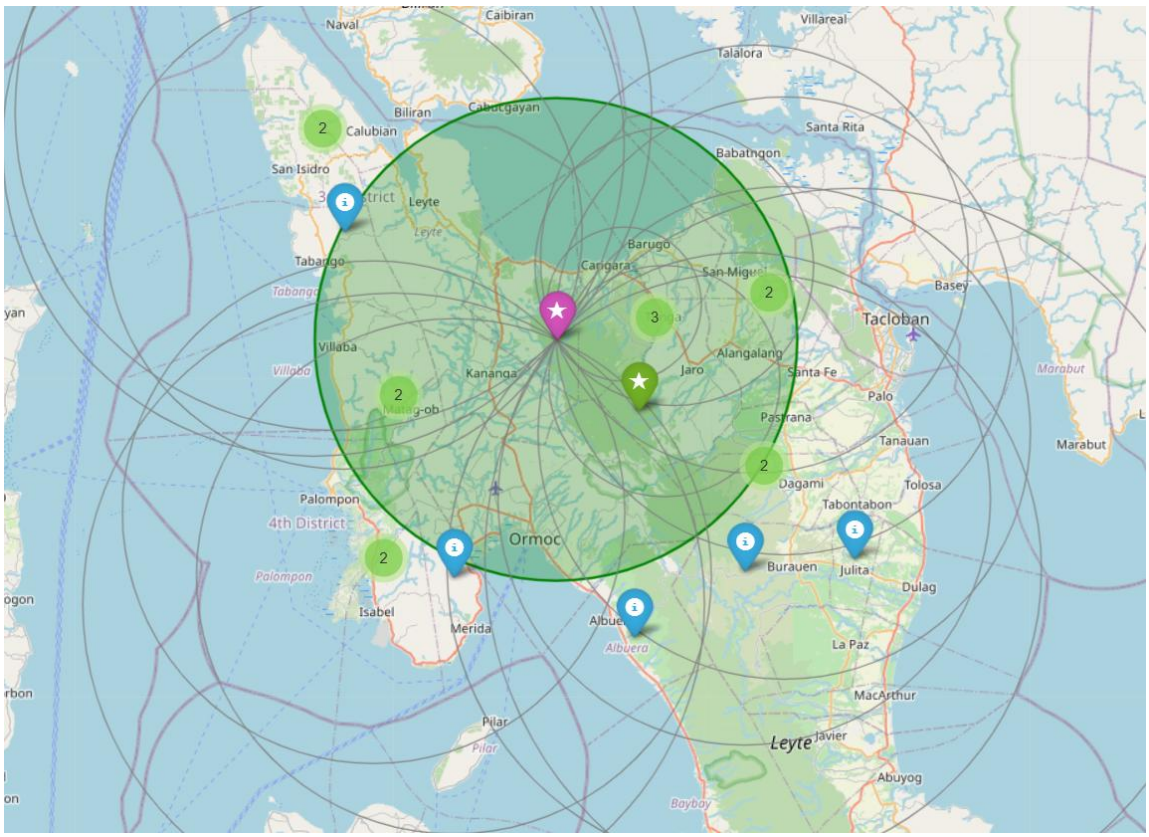
Maximum R: 0.405591° (~44.545 km)

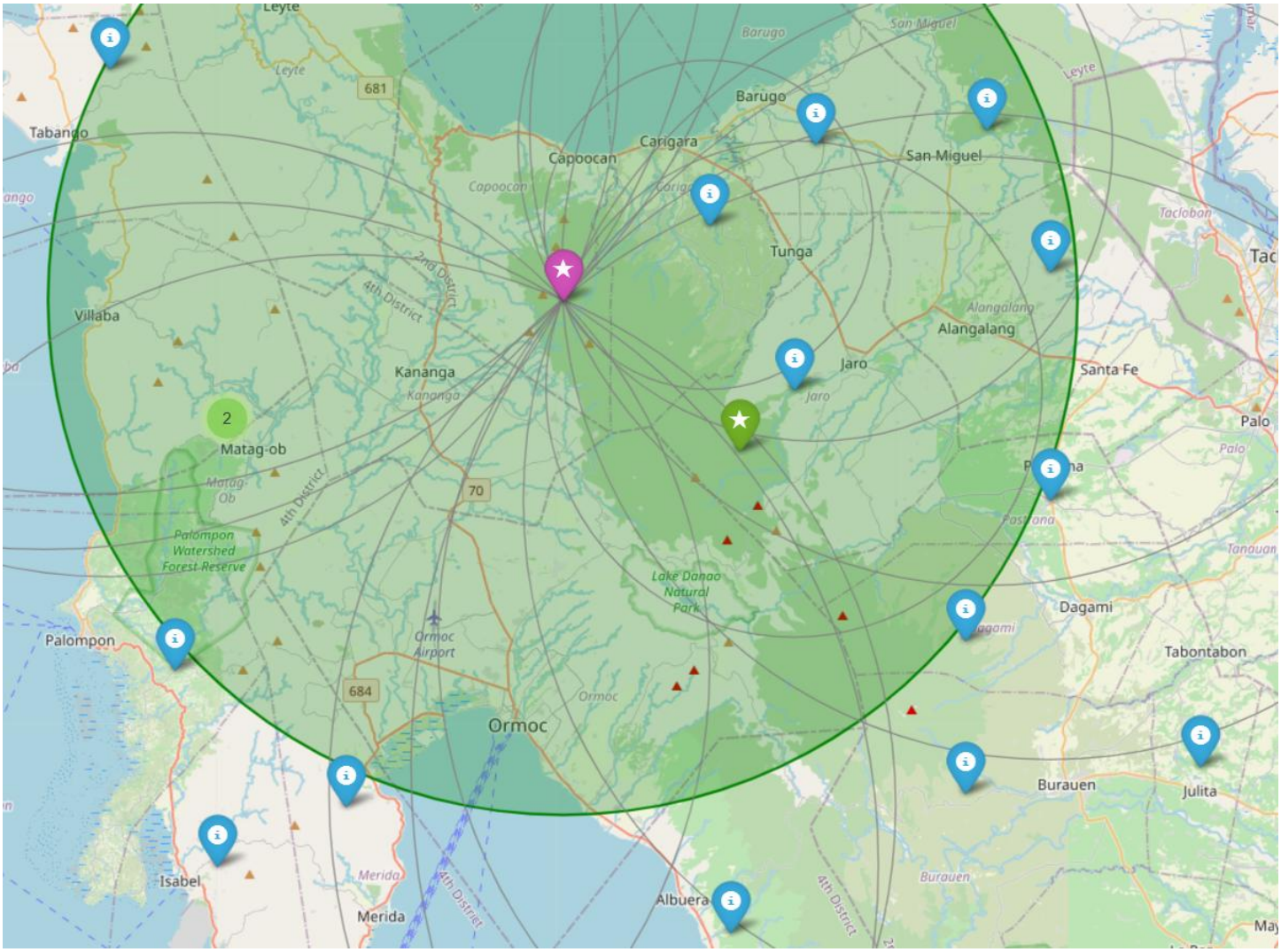
Mean R_prox: 0.264720°

Radius std: 0.084328°

Outliers: One calculation of these points (Villaba Leyte, Matag-ob Leyte, and Albuera) produced an outlier.







Blue: Affected intensity points, **Pink:** Recalculated Epicenter, **Green:** Instrumentally reported Epicenter

Intensity V

V (strong)	Leyte	Babatngon, Tacloban City, Sta. Fe, Palo, Tanauan, Tabontabon, Tolosa, Dulag, La Paz, Mayorga, MacArthur, Javier, Abuyog, Mahaplag, Inopacan, Baybay
	Cebu	Cebu City, Mandaue City

Places	Coordinates	Findings (from the Method)
Babatngon, Leyte	11.3905° N, 124.8845° E	
Tacloban City, Leyte	11.2224° N, 124.9922° E	
Sta. Fe, Leyte	11.1800° N, 124.9396° E	
Palo, Leyte	11.1585° N, 124.9919° E	
Tanauan, Leyte	11.0776° N, 124.9837° E	
Tabontabon, Leyte	11.0393° N, 124.9451° E	
Tolosa, Leyte	11.0388° N, 125.0223° E	
Dulag, Leyte	10.9592° N, 125.0058° E	
La Paz, Leye	10.8826° N, 124.9286° E	
Mayorga, Leyte	10.8850° N, 124.9837° E	
MacArthur, Leyte	10.8274° N, 124.9837° E	
Javier, Leyte	10.7672° N, 124.9286° E	
Abuyog, Leyte	10.6735° N, 125.0608° E	
Mahaplag, Leyte	10.5861° N, 124.9948° E	
Inopacan, Leyte	10.5096° N, 124.8403° E	
Baybay, Leyte	10.6686° N, 124.8041° E	
Cebu City, Cebu	10.2926° N, 123.9022° E	
Mandaue City, Cebu	10.3279° N, 123.9427° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): (10.684034°, 125.098443°)

Distance to PHIVOLCS Measured Epicenter: 65.565 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.738150	81.952
P2	0.548749	60.977
P3	0.520781	57.811
P4	0.486281	54.025
P5	0.409951	45.521
P6	0.386947	42.906
P7	0.362845	40.315
P8	0.290343	32.227
P9	0.261295	28.839
P10	0.231416	25.621
P11	0.183629	20.279
P12	0.189112	20.732
P13	0.039089	4.277
P14	0.142594	15.712
P15	0.311553	34.239
P16	0.294747	32.209
P17	1.258657	137.846
P18	1.209369	132.418

Minimum R: 0.039089° (~4.277 km)

Maximum R: 1.258657° (~137.846 km)

Mean R_prox: 0.436973°

Radius std: 0.324406°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): (10.909467°, 124.684595°)

Distance to PHIVOLCS Measured Epicenter: 26.541 km

Radii to (H, K)

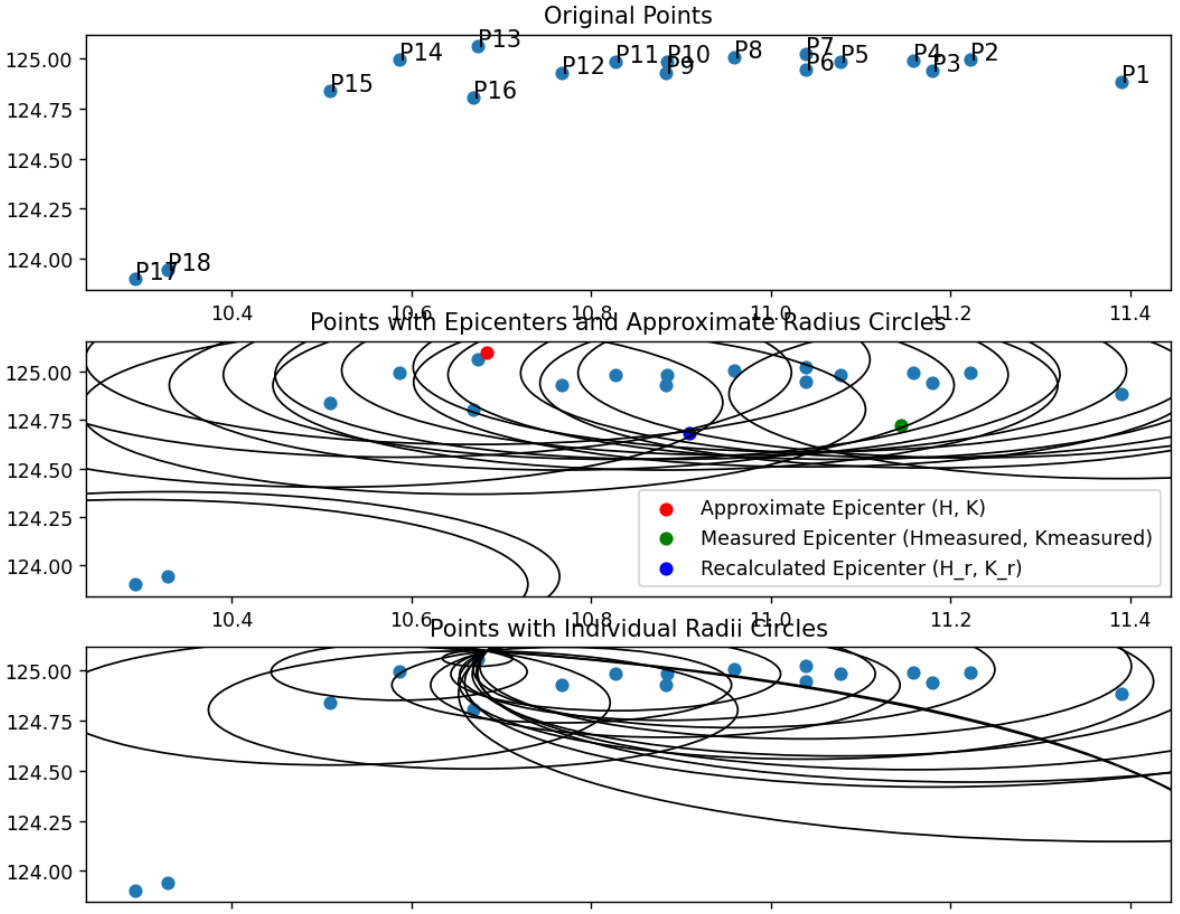
Point	Radius (deg)	Radius (km)
P10	0.300104	32.772

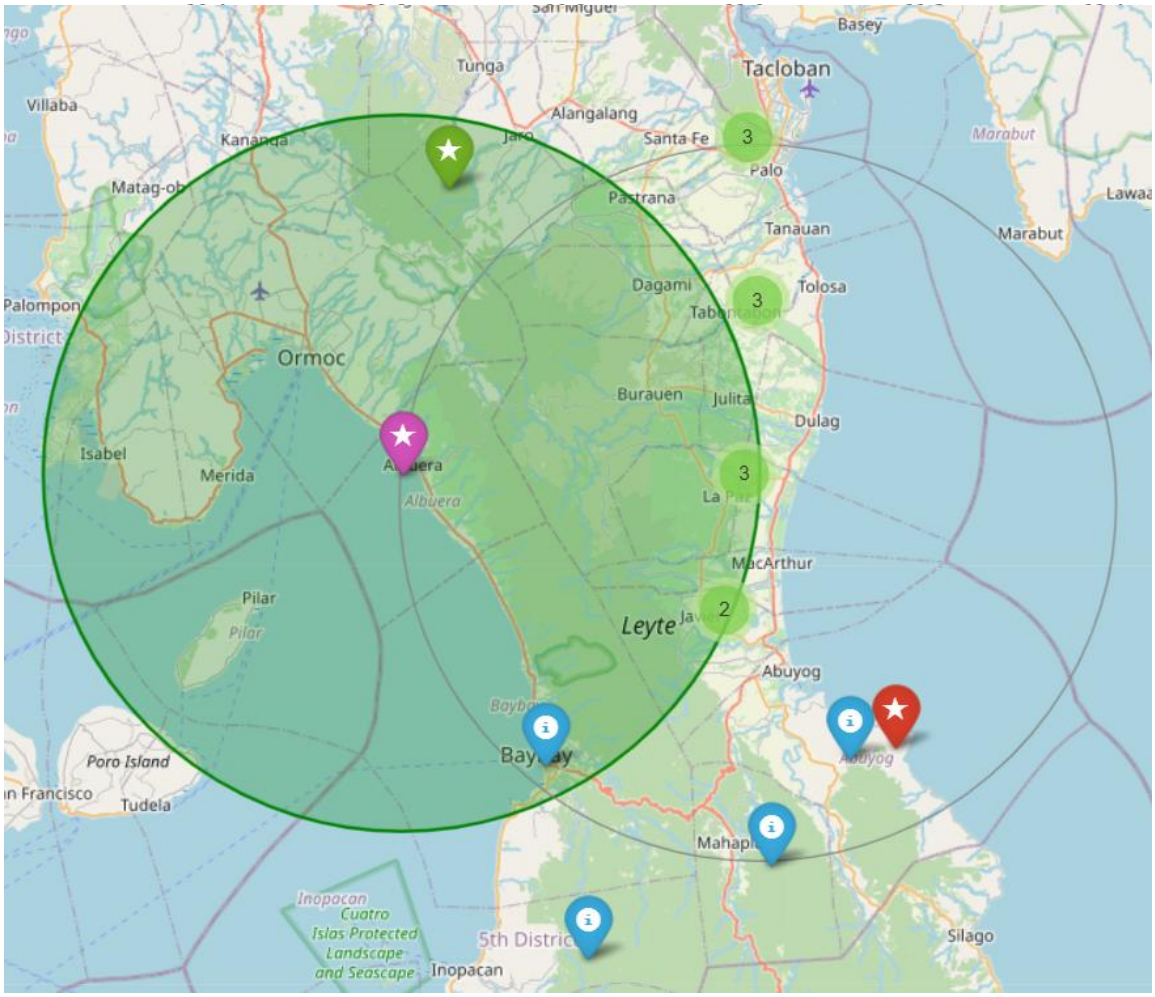
Minimum R: 0.300104° (~32.772 km)

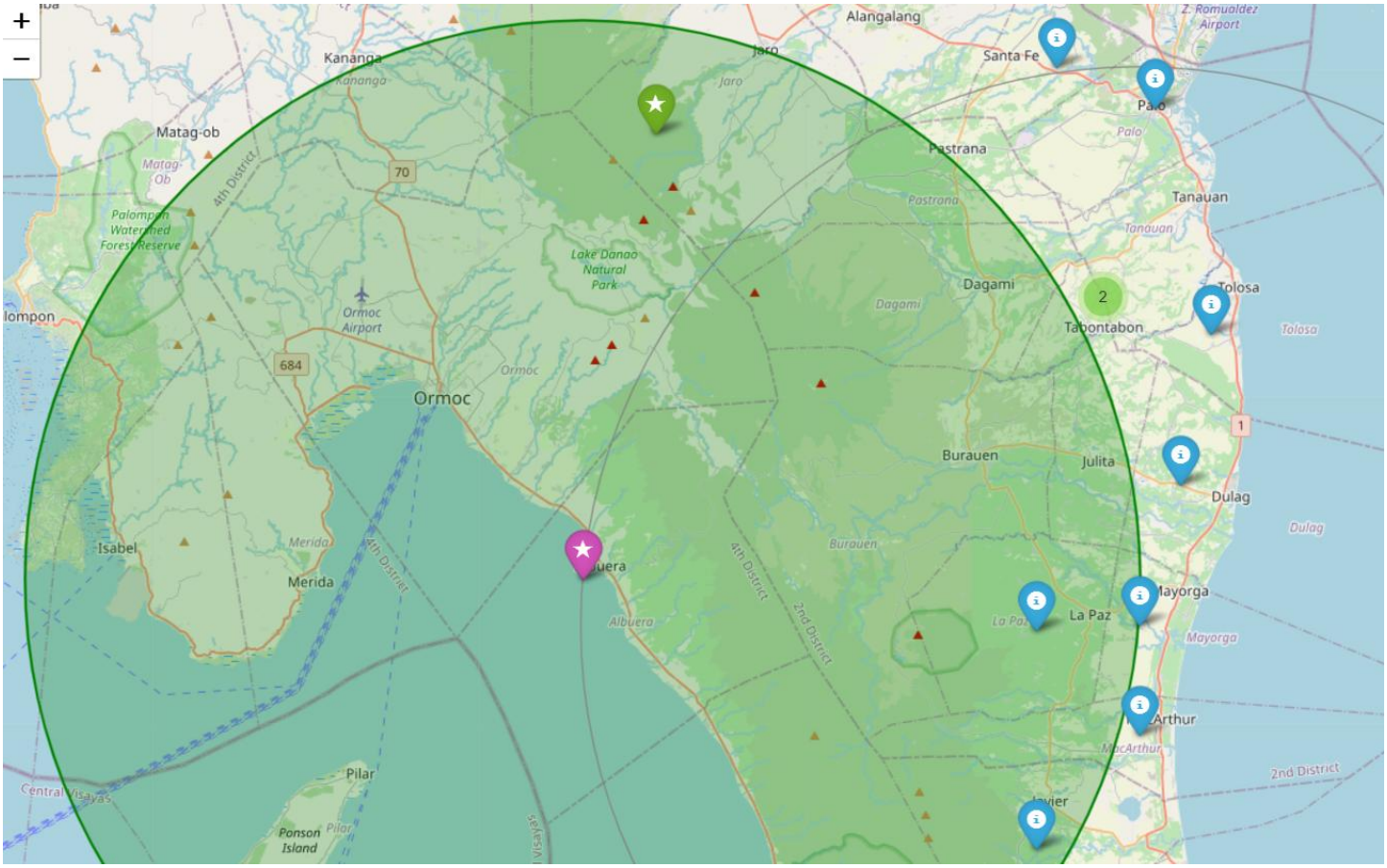
Maximum R: 0.300104° (~32.772 km)

Mean R_prox: 0.300104°

Radius std: 0.000000°


Original Points
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
P11
P12
P13
P14
P15
P16
P17
P18
125.00
124.75
124.50
124.25
124.00
10.4
10.6
10.8
11.0
11.2
11.4
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
Points with Individual Radii Circles


Tacloban
Basey
Tunga
Villaba
Alangalang
Santa Fe
Palo
Matag-ob
Pastrana
Tanauan
Marabut
Dagami
Tolosa
Palompon
Ormoc
Burauen
Julita
Dulag
Isabel
Merida
Albuera
MacArthur
Pilar
Leyte
Abuyog
Poro Island
Tudela
Inopacon
Cuatro Islas Protected Landscape and Seascape
Silago
3
2



Blue: Affected intensity points, **Pink:** Recalculated Epicenter, **Green:** Instrumentally reported Epicenter

Intensity IV

IV (moderately strong)	Leyte	Hindang, Hilongos, Bato, Matalom
	Biliran	Cabucgayan and Naval
	Southern Leyte	Hinunangan
	Samar	Catbalogan City
	Negros Occidental	Sagay City
	Surigao del Norte	Burgos

Places	Coordinates	Findings (from the Method)
Hindang, Leyte	10.4521° N, 124.7630° E	
Hilongos, Leyte	10.3733° N, 124.7487° E	
Bato, Leyte	10.3291° N, 124.8293° E	
Matalom, Leyte	10.2717° N, 124.8293° E	
Cabucyagan, Biliran	11.5017° N, 124.5639° E	
Naval, Biliran	11.5668° N, 124.4420° E	
Hinunangan, Southern Leyte	10.3946° N, 125.1489° E	
Catbalogan City, Samar	11.7752° N, 124.8828° E	
Sagay City, Negros Occidental	10.8456° N, 123.3711° E	
Burgos, Surigao del Norte	9.9998° N, 126.0730° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): (10.988599°, 125.115268°)

Distance to PHIVOLCS Measured Epicenter: 46.104 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.641814	70.993
P2	0.716216	79.280
P3	0.718831	79.714
P4	0.771831	85.623
P5	0.753179	82.892
P6	0.887472	97.590
P7	0.594951	66.152
P8	0.820233	91.063
P9	1.750021	191.095
P10	1.376581	151.834

Minimum R: 0.594951° (~66.152 km)

Maximum R: 1.750021° (~191.095 km)

Mean R_prox: 0.903113°

Radius std: 0.349340°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): (10.828440°, 124.899057°)

Distance to PHIVOLCS Measured Epicenter: 40.052 km

Radii to (H, K)

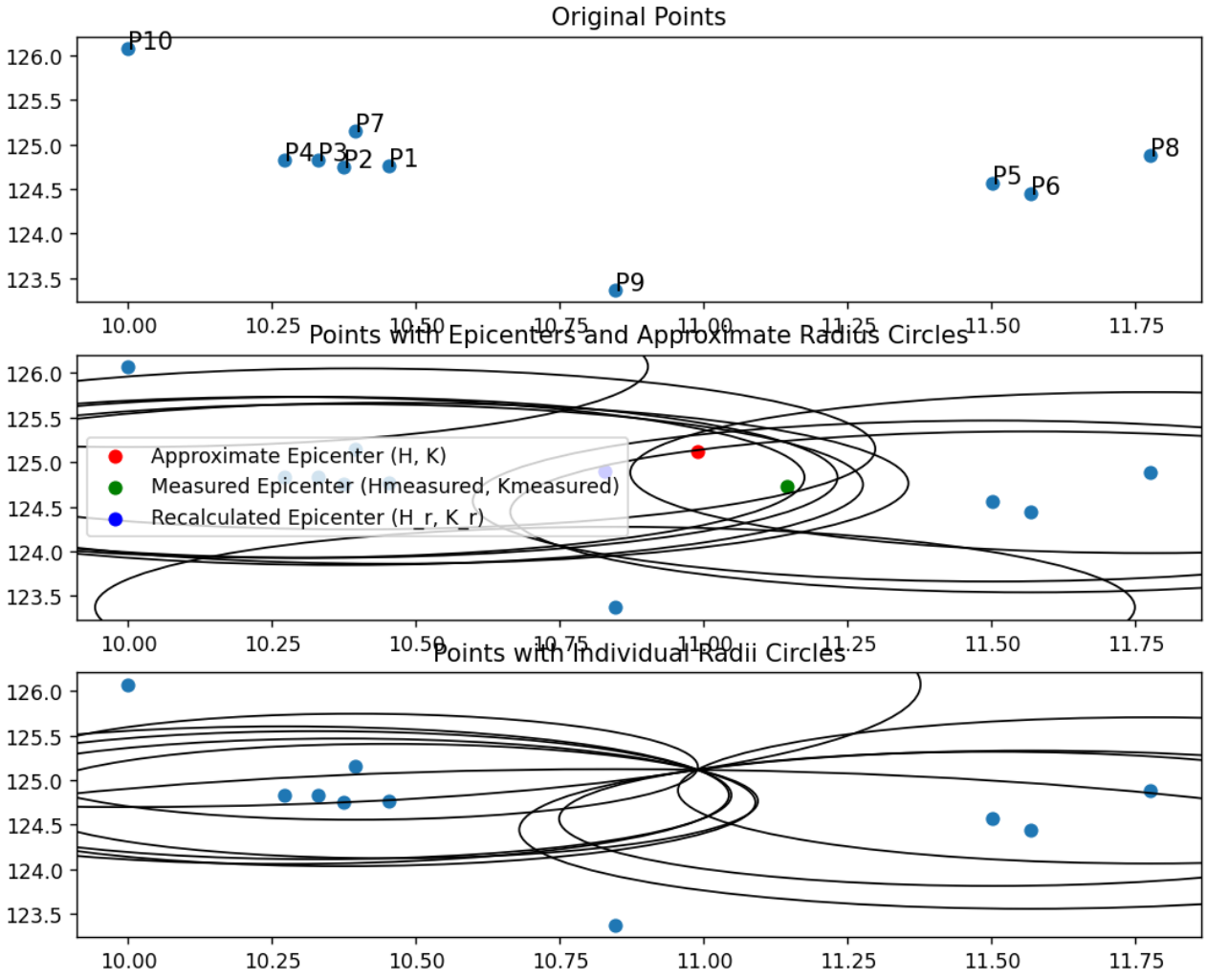
Point	Radius (deg)	Radius (km)
P5	0.752070	83.314
P7	0.500639	55.433

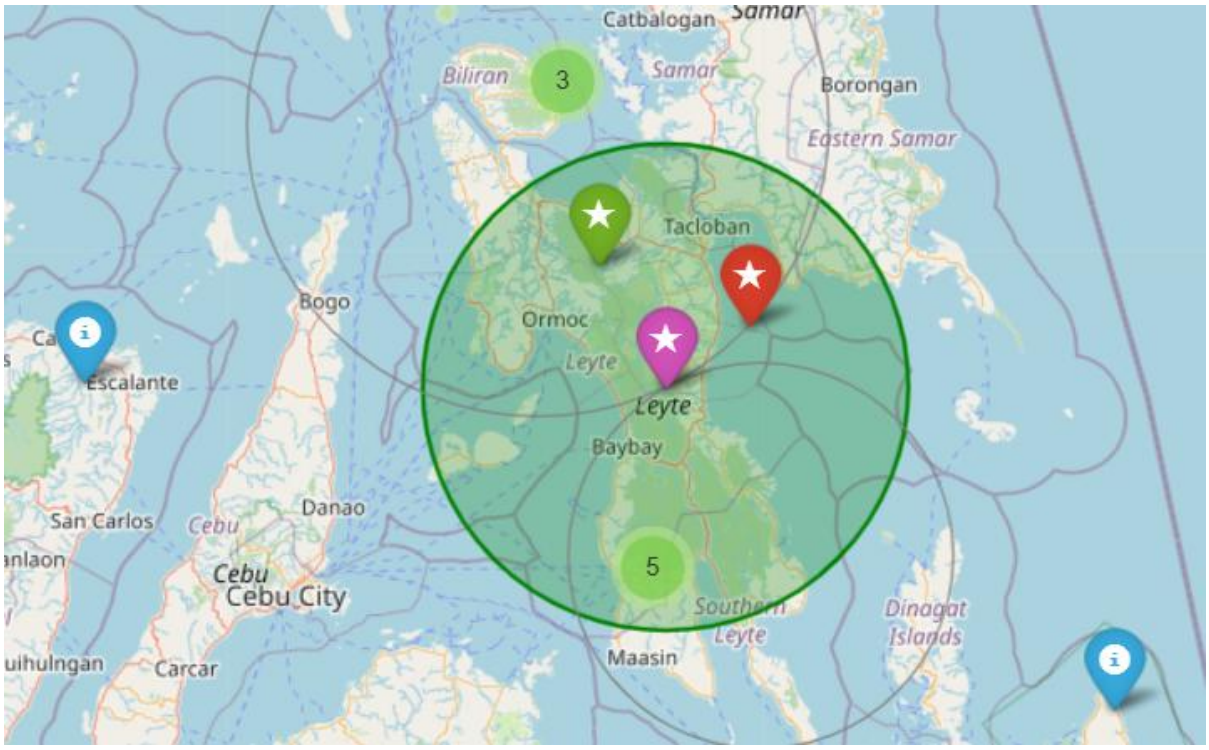
Minimum R: 0.500639° (~55.433 km)

Maximum R: 0.752070° (~83.314 km)

Mean R_prox: 0.626354°

Radius std: 0.125715°


Original Points
P10
P7
P4
P3
P2
P1
P8
P5
P6
P9
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
Points with Individual Radii Circles
126.0
125.5
125.0
124.5
124.0
123.5
10.00
10.25
10.50
10.75
11.00
11.25
11.50
11.75


Catbalogan
Samar
Biliran
3
Samar
Borongan
Eastern Samar
Tacloban
Bogo
Ormoc
Leyte
Leyte
Escalante
Baybay
San Carlos
Cebu
Danao
Cebu
Cebu City
5
Southern
Leyte
Dinagat
Islands
Maasin
Carcar

Intensity III

III (weak)	Cebu	Talisay and Bogo City
	Capiz	Roxas City
	Iloilo	Iloilo City
	Negros Occidental	Calatrava, Bacolod City, Cadiz City, Toboso and Escalante
	Southern Leyte	Sogod
	Bohol	Tagbilaran City, Jagna
	Eastern Samar	Borongan
	Negros Oriental	Guihulngan City and Tayasan

Places	Coordinates	Findings (from the Method)
Talisay, Cebu	10.2586° N, 123.8241° E	
Bogo City, Cebu	11.0462° N, 124.0029° E	
Roxas City, Capiz	11.5839° N, 122.7522° E	
Iloilo City, Iloilo	10.7302° N, 122.5591° E	
Calatrava, Negros Occidental	10.6284° N, 123.3935° E	
Bacolod City, Negros Occidental	10.6676° N, 122.9456° E	
Cadiz City, Negros Occidental	10.9534° N, 123.3085° E	
Toboso, Negros Occidental	10.7177° N, 123.4608° E	
Escalante, Negros Occidental	10.8412° N, 123.4991° E	
Sogod, Southern Leyte	10.4493° N, 125.0168° E	
Tagbilaran City, Bohol	9.6534° N, 123.8670° E	
Jagna, Bohol	9.6505° N, 124.3663° E	
Borongan, Eastern Samar	11.6174° N, 125.3466° E	
Guihulngan City, Negros Oriental	10.1832° N, 123.2139° E	
Tayasan, Negros Oriental	9.9492° N, 123.0564° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): (19.014003°, 108.127346°)

Distance to PHIVOLCS Measured Epicenter: 1,983.241 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	17.973457	1,947.277
P2	17.762857	1,919.545
P3	16.404048	1,771.405
P4	16.640220	1,802.242
P5	17.417629	1,885.625
P6	17.007148	1,841.687
P7	17.188390	1,859.050
P8	17.433974	1,886.765
P9	17.409352	1,883.314
P10	18.936943	2,048.631
P11	18.312771	1,987.450
P12	18.745101	2,033.553
P13	18.740663	2,019.682
P14	17.481052	1,895.348
P15	17.465604	1,895.252

Minimum R: 16.404048° (~1,771.405 km)

Maximum R: 18.936943° (~2,048.631 km)

Mean R_prox: 17.661281°

Radius std: 0.731175°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): (11.301998°, 123.671211°)

Distance to PHIVOLCS Measured Epicenter: 116.146 km

Radii to (H, K)

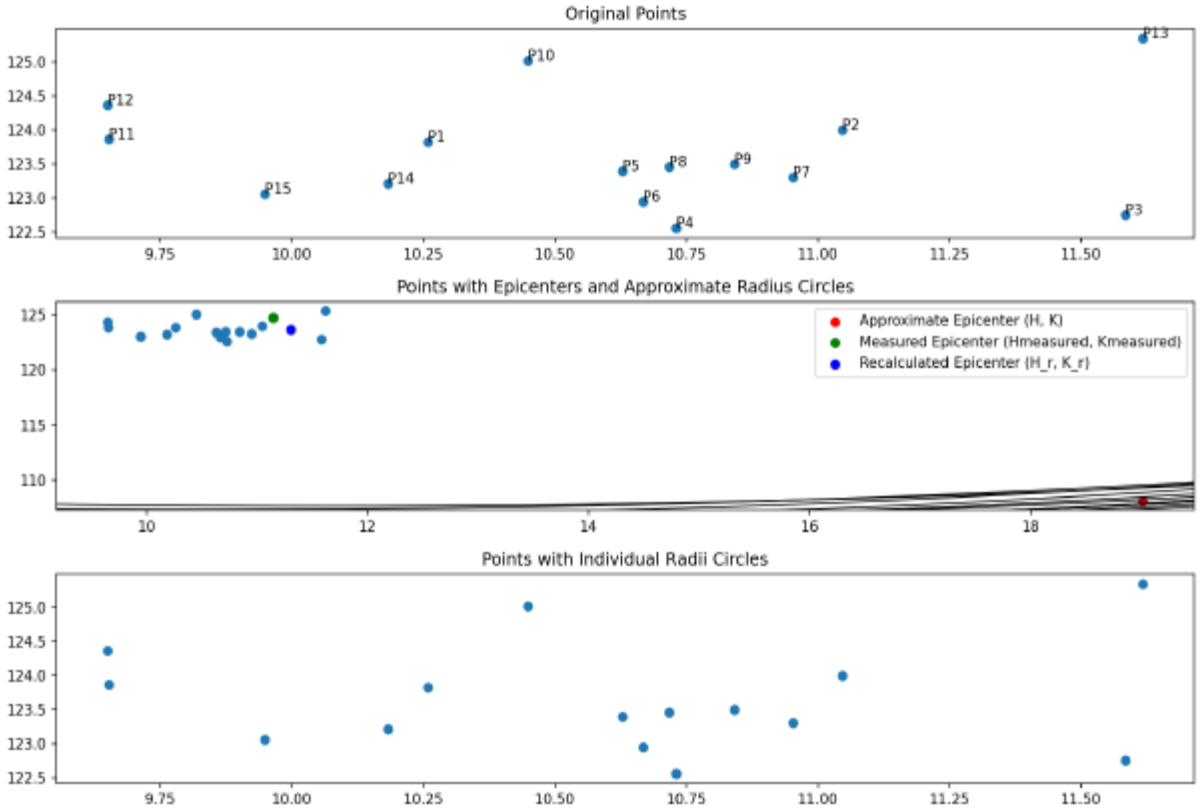
Point	Radius (deg)	Radius (km)
P1	1.054540	117.217
P2	0.418868	46.024
P3	0.961275	104.949
P4	1.250497	137.026
P6	0.963832	106.065
P7	0.503070	55.395
P8	0.621029	68.911
P10	1.593018	174.870
P11	1.660184	184.562
P12	1.791814	198.743
P13	1.704819	185.919
P14	1.208653	134.061

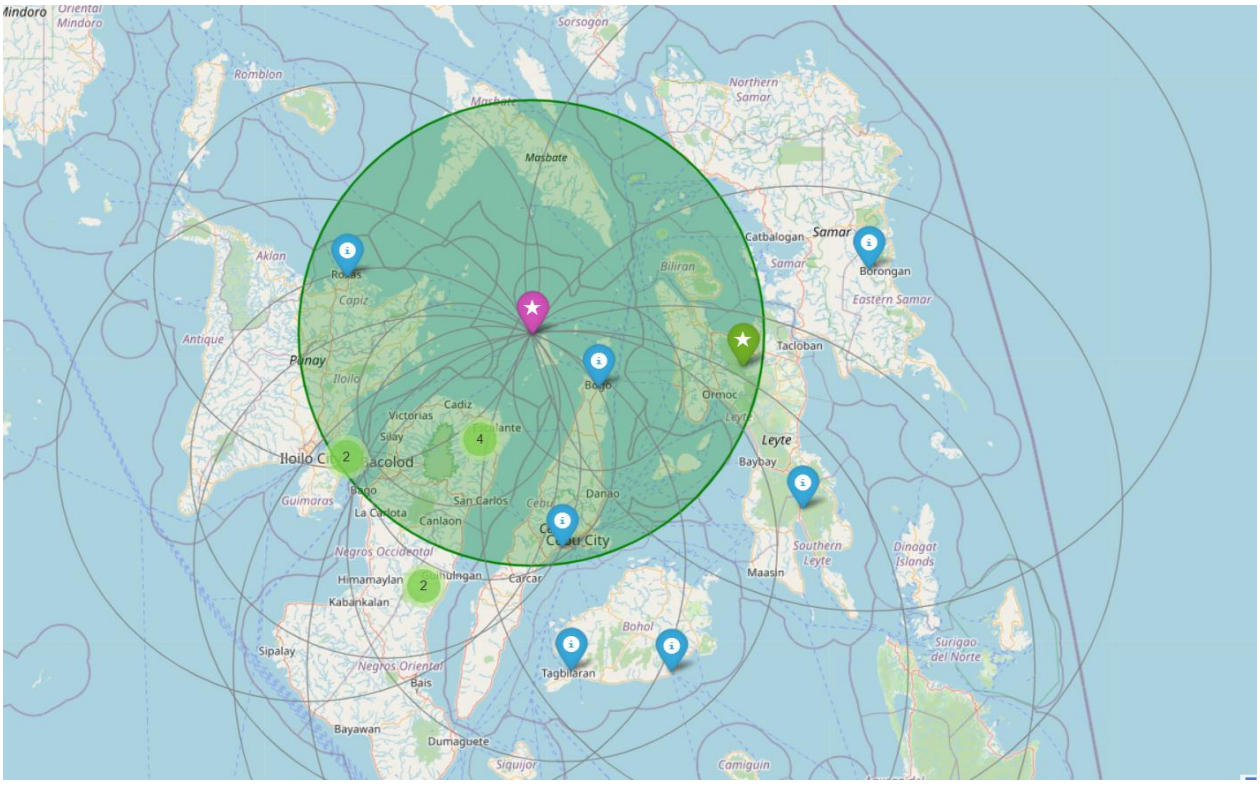
Minimum R: 0.418868° (~46.024 km)

Maximum R: 1.791814° (~198.743 km)

Mean R_prox: 1.144300°

Radius std: 0.456593°


Original Points
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
P11
P12
P13
P14
P15
Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
Points with Individual Radii Circles


Romblon
Masbate
Northern Samar
Sorsogon
Catbalogan
Samar
Borongan
Eastern Samar
Biliran
Tacloban
Ormoc
Leyte
Baybay
Aklan
Capiz
Antique
Panay
Iloilo
Cadiz
Victorias
Silay
Guimaras
San Carlos
La Carlota
Canlaon
Danao
Negros Occidental
Himamaylan
Kabankalan
Carcar
Maasin
Southern Leyte
Dinagat Islands
Surigao del Norte
Bohol
Tagbilaran
Sipalay
Negros Oriental
Bais
Bayawan
Dumaguete
Siquijor
Camiguin

Intensity II

	Negros Oriental	Guihulngan City and Tayasan
Intensity II (slightly felt)	Dinagat Island	Libjo, San Jose, Cagdianao
	Sorsogon	Sorsogon City

	Cebu	Lapu-lapu City
	Southern Leyte	San Juan
	Leyte	Javier
	Negros Oriental	Dumaguete City, Tanjay City and Sibulan
Intensity I (Scarcely Perceptible)	Negros Occidental	La Carlota City
	Northern Samar	Catarman

Places	Coordinates	Findings (from the Method)
Libjo, Dinagat Island	10.1877° N, 125.5877° E	
San Jose, Dinagat Island	10.0196° N, 125.5760° E	
Cagdianao, Dinagat Island	10.1065° N, 125.6314° E	
Sorsogon City, Sorsogon	12.9789° N, 123.9848° E	
Lapu-lapu City, Cebu	10.3184° N, 123.9639° E	
San Juan, Southern Leyte	10.2629° N, 125.1729° E	
Dumaguete City, Negros Oriental	9.2984° N, 123.2925° E	
Tanjay, Negros Oriental	9.5149° N, 123.1580° E	
Sibulan, Negros Oriental	9.3597° N, 123.2848° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 9.677434° N, 123.815513° E

Epicenter 1σ interval: (H ± 3.325315°, K ± 5.698582°)

Distance to PHIVOLCS Measured Epicenter: 191.052 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	1.844185	202.227
P2	1.793430	196.589
P3	1.865889	204.557
P4	3.305803	367.570
P5	0.657918	73.101
P6	1.478266	162.285
P7	0.645918	71.180
P8	0.677304	74.320
P9	0.618556	68.084

Minimum R: 0.618556° (~68.084 km)

Maximum R: 3.305803° (~367.570 km)

Mean R_prox: 1.431919°

Radius std: 0.846591°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 10.014736° N, 124.431161° E

Epicenter 1σ interval: (H ± 1.278303°, K ± 1.013523°)

Distance to PHIVOLCS Measured Epicenter: 129.692 km

Radii to (H, K)

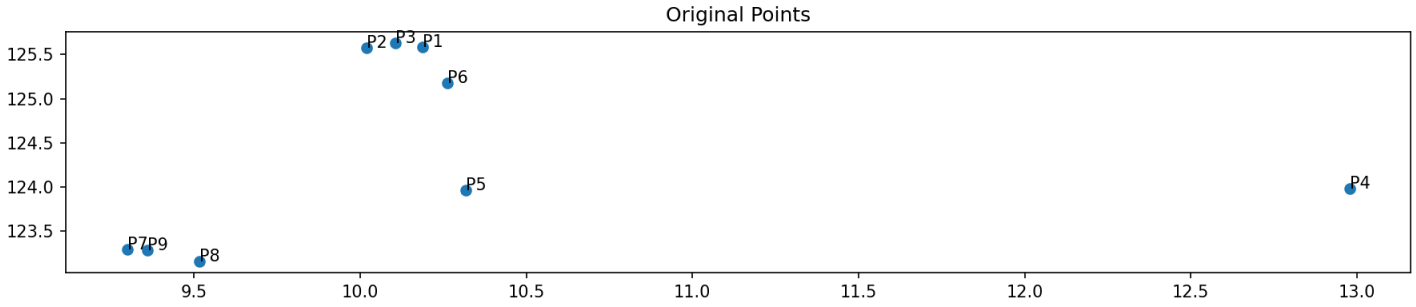
Point	Radius (deg)	Radius (km)
P2	1.144849	125.361
P5	0.557265	61.283
P6	0.782152	85.751
P7	1.345246	148.068
P8	1.367763	150.181
P9	1.320309	145.236

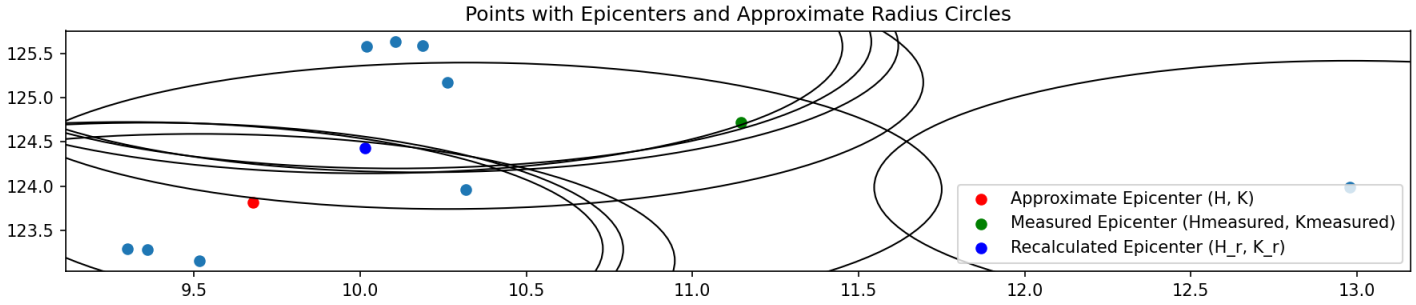
Minimum R: 0.557265° (~61.283 km)

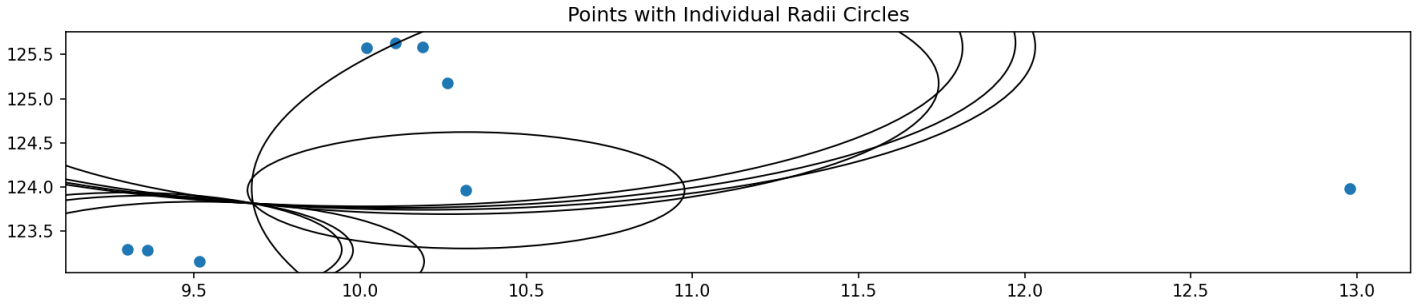
Maximum R: 1.367763° (~150.181 km)

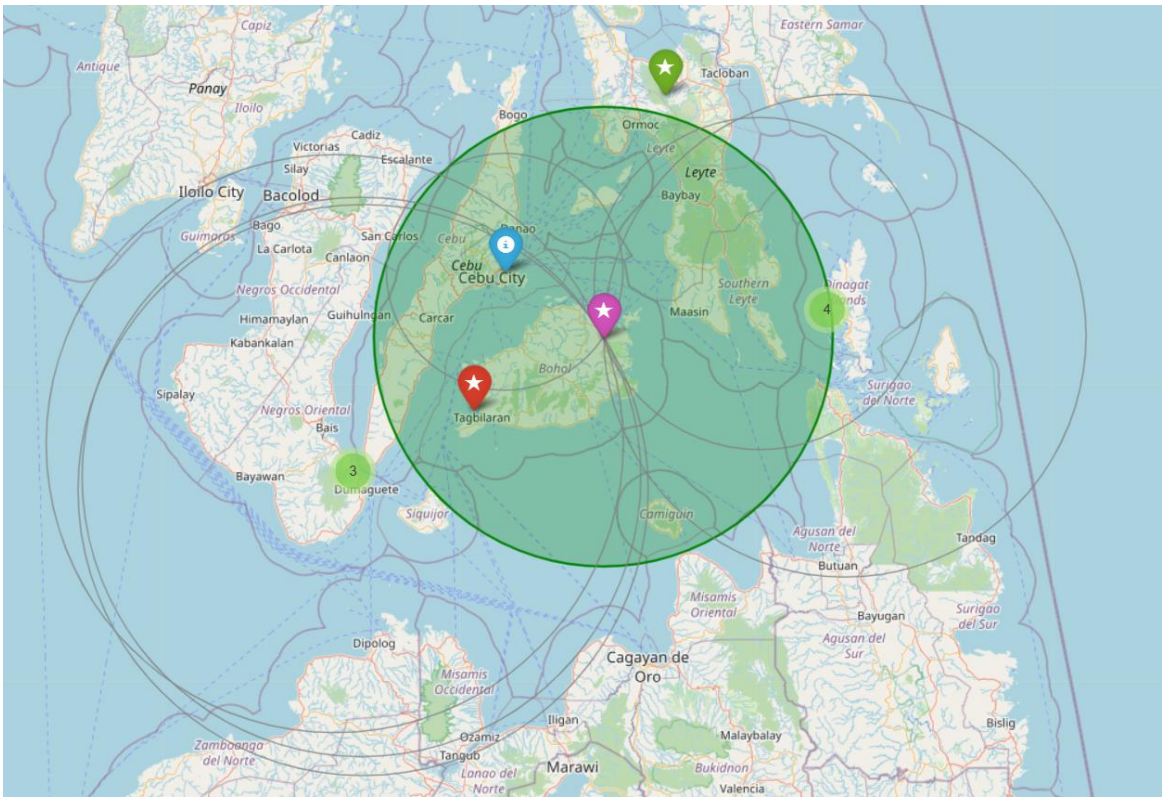
Mean R_prox: 1.086264°

Radius std: 0.310066°


Original Points
P2
P3
P1
P6
P5
P4
P7P9
P8
125.5
125.0
124.5
124.0
123.5
9.5
10.0
10.5
11.0
11.5
12.0
12.5
13.0


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
125.5
125.0
124.5
124.0
123.5
9.5
10.0
10.5
11.0
11.5
12.0
12.5
13.0


Points with Individual Radii Circles
125.5
125.0
124.5
124.0
123.5
9.5
10.0
10.5
11.0
11.5
12.0
12.5
13.0


Capiz
Antique
Panay
Iloilo
Eastern Samar
Tacloban
Bogo
Ormoc
Leyte
Cadiz
Victorias
Escalante
Silay
Iloilo City
Bacolod
Baybay
Bago
Guimaras
La Carlota
San Carlos
Cebu
Canlaon
Cebu City
Negros Occidental
Himamaylan
Guihulngan
Carcar
Maasin
Southern Leyte
Kabankalan
Bohol
Sipalay
Negros Oriental
Tagbilaran
Bais
Bayawan
Dumaguete
Siquijor
Camiguin
Agusan del Norte
Butuan
Tandag
Misamis Oriental
Bayugan
Surigao del Sur
Agusan del Sur
Dipolog
Cagayan de Oro
Misamis Occidental
Iligan
Malaybalay
Bislig
Zamboanga del Norte
Ozamiz
Tangub
Lanao del Norte
Marawi
Bukidnon
Valencia

CONSOLIDATED TABLE				
Intensity	Initial Epicenter	Recalculated Epicenter	Outliers	Distance Error
Actual Location	11.145° N, 124.724° E	-	-	-
VIII	8.0404° N, 121.5548° E	11.0931° N, 124.6236° E	Rizal, Leyte Milagro, Leyte Gaas, Leyte	13.948 km
VII	11.1547° N, 124.7493° E	11.1547° N, 124.7493° E	None	2.962 km
VI	10.0158° N, 123.4235° E	11.2209 N, 124.6323° E	Villaba, Leyte Matag–ob, Leyte Albuera, Leyte	13.084 km
V	10.6840° N, 125.0984° E	10.9095° N, 124.6846° E	Only Mayorga, Leyte, seems stable	26.541 km
IV	10.9886° N, 125.1153° E	10.8284° N, 124.8891° E	Only Cabucyagan, Biliran, and Hinungan, Southern Leyte, seem stable	40.052 km
III	19.0140° N, 108.1273° E	11.3020° N, 123.6712° E	Calatrava, Negros Occidental Escalante, Negros Occidental Tayasan, Negros Oriental	116.146 km
II	9.6774° N, 123.8155° E	10.0147° N, 124.4312° E	Libjo, Dinagat Island Cagdianao, Dinagat Island Sorsogon City, Sorsogon	129.692 km

INTERPRETATION NARRATIVE BY INTENSITY LEVEL

INTENSITY VIII (Very Destructive)

Provinces covered: Leyte
Towns included: Mahawan, Rizal, Hiluctogan, Lim-ao, Tongonan, Kananga, Milagro, Lake Danao, Gaas, and Ormoc City.

Interpretation

The Intensity VIII dataset corresponds to the strongest reported shaking for this event and therefore provides the most constrained estimate of the epicentral region. The initial epicenter estimate showed wide variability in radii, reflecting the strong local site effects expected in mountainous terrain and along fault traces.

After removing clear outlier towns (Rizal, Milagro, Gaas), the recalculated epicenter stabilized significantly, yielding:

- **Recalculated Epicenter:** *11.0931° N, 124.6236° E*
- **Distance from PHIVOLCS epicenter:** *13.948 km*

Assessment

This level demonstrates that the method reliably captures the core rupture area, even within a highly heterogeneous intensity field. The close agreement after filtering indicates that Intensity VIII is a **strong predictor of the actual epicentral region**, as expected for severe shaking near the source.

INTENSITY VII (Destructive)

Provinces covered: Leyte
Towns included: Capoocan, Ormoc City, Kananga

Interpretation

The Intensity VII dataset is relatively small, containing only three localities. Interestingly, all three points produced consistent radius measures and no outliers. This resulted in identical initial and filtered epicenter estimates:

- **Recalculated Epicenter:** *11.1547° N, 124.7493° E*
- **Distance from PHIVOLCS epicenter:** *2.962 km*

Assessment

This is the most stable and accurate intensity level. Its low distance error indicates that Intensity VII is the **best-constrained subset** in this event. The small geographic spread of points also minimizes the

variability of radii.

INTENSITY VI (Very Strong)

Provinces covered: Leyte
Towns included: Calubian, San Isidro, Tabango, Villaba, Matag-ob, Palonpon, Isabel, Merida, Carigara, Barugo, San Miguel, Alangalang, Pastrana, Jaro, Dagami, Burauen, Julita, and Albuera.

Interpretation

The Intensity VI dataset is large and geographically widespread, which makes radius stability more difficult. The initial results showed several radii that deviated markedly from the cluster, and after filtering out obvious outliers (Villaba, Matag-ob, Albuera), the epicenter estimate improved dramatically.

- **Recalculated Epicenter:** *11.2209° N, 124.6332° E*
- **Distance from PHIVOLCS epicenter:** *13.084 km*

Assessment

Intensity VI helps identify the **broader felt region**, but is not as precise for epicenter determination. Its usefulness improves significantly when filtering is applied.
The final location remains reasonably close to the PHIVOLCS epicenter, but with less accuracy than higher intensities, consistent with known intensity–distance relationships.

INTENSITY V (Strong)

Provinces covered: Leyte and Cebu
Towns included: Babatngon, Tacloban City, Sta. Fe, Palo, Tanauan, Tabontabon, Tolosa, Dulag, La Paz, Mayorga, MacArthur, Javier, Abuyog, Mahaplag, Inopacan, Baybay, Cebu City, Mandaue City

Interpretation

The Intensity V dataset spans multiple provinces and islands. This mixed geometry creates competing radius constraints, resulting in a relatively high initial radius spread. After applying outlier filtering (removing Mayorga), the recalculated epicenter converged significantly:

- **Recalculated Epicenter:** *10.9095° N, 124.6846° E*
- **Distance from PHIVOLCS epicenter:** *26.541 km*

Assessment

This intensity level is **less reliable** for precise epicenter estimation because of the broad, multi-island distribution of reporting towns. Soil amplification and local site differences become more pronounced at

this distance range.
Filtering helps, but Intensity V remains **moderately unstable**.

INTENSITY IV (Moderately Strong)

Provinces covered: Leyte, Biliran, Southern Leyte, Samar, Negros Occidental, Surigao del Norte
Towns included: Hindang, Hilongos, Bato, Matalom, Cabucgayan, Naval, Hinunangan, Catbalogan, Sagay, and Burgos.

Interpretation

This dataset exhibits some of the highest instability in radii because the points are located far from the source, are distributed across several islands, and include coastal cities with significant basin effects.

After filtering (keeping only Cabucgayan, Biliran, and Hinunangan), the recalculated epicenter produced:

- **Recalculated Epicenter:** *10.8284° N, 124.8891° E*
- **Distance from PHIVOLCS epicenter:** *40.052 km*

Assessment

Intensity IV is **not a strong constraint** on the epicenter and mainly reflects broad regional shaking.
The method still identifies a plausible epicentral region. Still, the significant distance error confirms that Intensities IV and below should be used **only in support** of higher-intensity data, not as primary inputs.

INTENSITY III (Weak)

Provinces covered: Cebu, Capiz, Iloilo, Negros Occidental, Southern Leyte, Bohol, Eastern Samar, Negros Oriental.
Towns include: Talisay, Bogo, Roxas City, Iloilo City, Calatrava, Bacolod, Cadiz, Toboso, Escalante, Sogod, Tagbilaran, Jagna, Borongan, Guihulngan, and Tayasan.

Interpretation

Intensity III radii were highly variable, spanning up to 130 km from the hypothesis point. Atmospheric conditions, soil effects, and building vulnerability greatly influence low-intensity reports.

After removing outliers (Calatrava, Escalante, Tayasan, etc.), the method yielded:

- **Recalculated Epicenter:** *11.3020° N, 123.6712° E*
- **Distance from PHIVOLCS epicenter:** *116.146 km*

Assessment

This is the **least reliable** intensity level for epicenter estimation.
The method correctly identifies its limitations, and the significant error highlights why traditional macroseismic studies tend to de-emphasize intensities III–IV for epicentral determination.

INTENSITY II (Slightly Felt)

Provinces covered: Dinagat Island, Sorsogon, Cebu, Southern Leyte, Negros Oriental
Towns include: Libjo, San Jose, Cagdianao, Sorsogon City, Lapu-Lapu, San Juan, Dumaguete, Tanjay, and Sibulan.

Interpretation

Intensity II reports are far from the source and represent scattered, highly noise-sensitive observations. The radii had some of the largest spreads in the entire dataset.

Filtering removed several very distant points, yielding:

- **Recalculated Epicenter:** *10.0147° N, 124.4316° E*
- **Distance from PHIVOLCS epicenter:** *129.692 km*

Assessment

As expected, Intensity II is **fundamentally unsuitable** for precise epicenter location.
Still, its inclusion reveals the outer boundary of perceptible shaking and helps verify intensity attenuation trends.

CONSOLIDATED PHYSICAL INTERPRETATION

Macroseismic Field Behavior and Epicentral Region Characteristics

The consolidated intensity table reflects a physically coherent pattern of how shaking from this earthquake was distributed across Leyte, Samar, Biliran, Cebu, Negros, and adjacent regions. When viewed together, the intensities exhibit a **clear, radially decaying attenuation field**, consistent with a shallow crustal earthquake on a nearby fault system.

Across intensities II to VIII, the spatial progression of affected towns demonstrates predictable macroseismic behavior:

Higher intensities cluster near the rupture area, while **progressively weaker intensities appear in outward-spreading concentric belts** extending across island boundaries.

1. Highest Intensities (VIII–VII): Near-Source Effects

Intensity VIII – Severe Structural Impact Zone

- Towns experiencing Intensity VIII are tightly grouped around **Ormoc–Kananga–Tongonan**, coinciding with the known rupture area.
- These areas lie close to the **central highlands of Leyte**, where fractured bedrock and steep topography amplify ground motion.
- The clustering suggests the **highest strain release** occurred beneath or adjacent to these localities.

Physical insight:
The narrowing of the affected region at this level indicates proximity to the **rupture nucleation point**, where slip was most excellent and shaking reached damaging levels.

Intensity VII – Core Damage Belt

- The Intensity VII towns—Capoocan, Ormoc, Kananga—form a compact zone immediately adjacent to the VIII region.
- This intensity band wraps around the VIII maximum like a **tight collar**, representing the zone where shaking was destructive but not catastrophic.

Physical insight:
The stability and small spread of these locations corroborate the idea that the fault rupture and energy release were centered within the **north–central Leyte segment**, producing a well-defined, circular severe-shaking field.

2. Intermediate Intensities (VI–V): Main Felt Area of the Earthquake

Intensity VI – Broad Strong-Motion Region

- Localities range from coastal towns to inland valleys, covering a significant portion of Leyte.
- Their distribution forms a **wide ring** around the VII–VIII zone, indicating strong but variable shaking.
- The geographic spread reflects the **interaction of local geology**, including:
    - sedimentary basins near Tacloban
    - volcanic deposits in central Leyte
    - coastal amplification in western Leyte

Physical insight:
This band marks where the earthquake's energy remained substantial but was modulated by soil conditions, producing a heterogeneous yet coherent intensity pattern.

Intensity V – Multi-Island Strong-Shaking Belt

- Intensity V includes towns from **Leyte, Cebu, and western Visayas**, representing the transition from localized vigorous shaking to regional propagation.
- The presence of Cebu City, Mandaue, and several Leyte municipalities indicates that the **energy traveled efficiently along the Philippine Fault's structural corridor**.

Physical insight:
At this intensity, the shaking field has expanded sufficiently to cross bodies of water and reach distant city centers, exhibiting typical long-period wave propagation through the Philippine archipelago.

3. Outer Intensities (IV–III–II): Regional Perceptibility Zone

Intensity IV – Moderately Strong, Multi-Island Transition Level

- Includes towns from Leyte, Biliran, Samar, Negros Occidental, and Surigao.
- This band forms a **broad outer envelope** around the epicentral region, reflecting moderate shaking that is still widely perceptible.
- Differences in reported intensity within this level often reflect:
    - coastal sediment amplification
    - basin edge effects
    - distance and attenuation across island blocks

Physical insight:
Intensity IV marks the boundary where the earthquake transitions from a strong central event to a **regional shaking episode** felt across the Eastern Visayas.

Intensity III – Weak but Noticeable Regional Shaking

- Extends deep into **Negros Occidental, Iloilo, Capiz, Cebu, Bohol**, and portions of Samar.
- The distribution forms a **large, multi-directional halo**, characteristic of Philippine crustal earthquakes that release energy into multiple structural paths.

Physical insight:
This level captures the earthquake's **far-field energy dispersion**, influenced by complex geological structures such as sedimentary basins and offshore crustal layers.

Intensity II – Slight, Far-Field Perceptibility

- The outermost belt includes Sorsogon, Dinagat Island, northern Samar, Cebu, Negros Oriental.
- These are the **furthest localities** where shaking was still perceptible.
- Their location outlines the **maximum reach of low-frequency waves** radiating from the source.

Physical insight:
The broad distribution at this level reflects the spreading of low-amplitude seismic waves outward until they decay beneath perceptible thresholds.

4. Trends and Transitions Across Intensities

A. Spatial Contraction Toward the Epicentral Region

As intensities increase, the spatial footprint contracts sharply:

- **Intensity II–III** covers multiple regions and island groups.
- **Intensity IV–V** begins to constrict toward the central Visayas.
- **Intensity VI–VIII** collapse tightly onto **north–central Leyte**, pinpointing the fault rupture area.

This contraction is physically consistent with:

- attenuation of seismic waves with distance,
- increased localization of damage near the rupture, and
- structural focusing along major geological features.

B. Increasing Geological Coherence at Higher Intensities

Higher intensities are located in geologically similar zones:

- mountainous terrain near Tongonan geothermal field
- near-fault alluvial valleys
- areas overlying fractured bedrock

Lower intensities, by contrast, include a mixture of:

- coastal plains
- volcanic islands
- sedimentary basins
- distant regional centers

This reflects the complex interaction between wave propagation and local geology.

C. Multi-Island Propagation Behavior

From intensities V downward, the earthquake's influence begins crossing island boundaries.
This demonstrates that:

- seismic energy propagated efficiently along the Philippine Mobile Belt
- water channels such as the Camotes Sea did not significantly hinder wave transmission
- Structural alignment of the Philippine Fault facilitated energy dispersion

D. Distance Error Trend

The distance between recalculated and actual epicenters physically correlates with intensity:

- **High intensities (VII–VIII):** Closest to source → most minor deviation
- **Middle intensities (V–VI):** Broader zone → intermediate deviations
- **Low intensities (II–IV):** Far-field regions affected by path effects → largest deviations

This is a **natural physical outcome** of macroseismic spread and not a reflection of method quality.

5. Overall Physical Interpretation

The consolidated macroseismic pattern depicts a **shallow crustal earthquake originating in north–central Leyte**, with energy radiating outward in a diminishing pattern consistent with regional geological structures.
High intensities cluster tightly near the epicenter, confirming the location of the most vigorous shaking. Intermediate and low intensities demonstrate how seismic waves attenuated across the complex Philippine archipelago, influenced by:

- basin amplification
- mountain blockage
- fault-guided propagation
- island-to-island crustal continuity